

# E-COMMERCE & PROFIT ANALYSIS

## PROJECT DOCUMENTATION

Business analysis •  
Python EDA • Power  
BI • DAX

Sample - Superstore •  
2014–2017

| AUTHOR        | ROLE                  | REPORT TYPE                     |
|---------------|-----------------------|---------------------------------|
| Aditaya Singh | Aspiring Data Analyst | Portfolio Project Documentation |

**Professional portfolio edition** — structured to communicate the analytical process, the eight Python business questions, dashboard metrics, DAX logic, insights, and business recommendations.

## 01. Executive Snapshot

This project analyzes e-commerce transaction data to understand sales performance, profitability, product/category performance, customer segments, and business opportunities. Python is used for exploratory analysis, while Power BI and DAX are used to build the interactive reporting layer.

| TOTAL SALES | TOTAL PROFIT | ORDERS | PROFIT MARGIN | AOV      | YOY GROWTH |
|-------------|--------------|--------|---------------|----------|------------|
| \$2.30M     | \$286.40K    | 5,009  | 12.47%        | \$458.61 | 20.36%     |

## 02. Project Overview

The project focuses on converting transaction-level e-commerce data into business-facing analysis. The final output combines Python exploratory analysis with a two-page Power BI report covering sales performance, profitability, customer segments, categories, sub-categories, and business insights.

## 03. Business Objective

- Analyze overall sales and profit performance.
- Identify sales trends over time.
- Compare category and sub-category performance.
- Analyze customer segment performance.
- Identify high-profit and low-profit areas.
- Generate actionable business recommendations.

## 04. Dataset & Analytical Scope

**Dataset:** Sample - Superstore (2014–2017)

Core fields include Order ID, Order Date, Customer, Customer Segment, Product, Category, Sub-Category, Sales, Quantity, Discount, Profit, Region, State, City, and Ship Mode.

**Primary tools:** Python • Pandas • Matplotlib • Power BI • DAX • CSV

## 05. Project Workflow

|  |                                  |
|--|----------------------------------|
|  | Raw E-commerce Data              |
|  | Data Cleaning & Preparation      |
|  | Python Exploratory Data Analysis |
|  | Business Analysis                |
|  | Power BI Dashboard Development   |
|  | DAX Calculations                 |
|  | Key Business Insights            |
|  | Business Recommendations         |

## 06. Data Preparation

- Checking missing/null values.
- Checking duplicate records.
- Reviewing data types.
- Standardizing date-related fields.
- Validating important numerical columns.
- Preparing the dataset for analysis and visualization.

The dataset was checked for missing values and duplicate records before analysis.

## 07. Python Exploratory Data Analysis — 8 Business Questions

**This section is included directly from the project notebook analysis.** It documents the business questions, the analytical focus, and the reported results.

### Q1 — Monthly Sales Analysis

**Business question:** Calculate monthly sales and identify the highest- and lowest-sales months.

**Reported result:** Highest: November — \$352,461.09

Lowest: February — \$59,751.26

## Q2 — Sales by Category

**Business question:** Analyze sales by product category and identify the highest- and lowest-sales categories.

**Reported result:** Highest: Technology — \$836,154.10  
Lowest: Office Supplies — \$719,046.99

## Q3 — Sales by Sub-Category

**Business question:** Analyze sales by sub-category and identify the highest- and lowest-sales sub-categories.

**Reported result:** Highest: Phones — \$330,007.10  
Lowest: Fasteners — \$3,024.25

## Q4 — Monthly Profit Analysis

**Business question:** Analyze monthly profit and identify the month with the highest profit.

**Reported result:** Highest: December — \$43,369.23

## Q5 — Profit by Category and Sub-Category

**Business question:** Analyze profit across category and sub-category combinations.

**Reported result:** Highest individual sub-category profit: Copiers — \$55,617.90  
Example of negative profit: Tables — -\$17,725.59

## Q6 — Sales & Profit by Customer Segment

**Business question:** Analyze sales and profit across customer segments.

**Reported result:** Consumer: \$1,161,401.34 sales / \$134,119.33 profit  
Corporate: \$706,146.44 / \$91,979.45  
Home Office: \$429,653.29 / \$60,299.01

## Q7 — Sales-to-Profit Ratio

**Business question:** Analyze the relationship between total sales and total profit.

**Reported result:** Sales-to-Profit Ratio: 8.02

## Q8 — Sales by State

**Business question:** Identify the state with the highest sales.

**Reported result:** California — \$457,687.68

# 08. Power BI Dashboard

## Page 1 — Sales Performance Overview

- Total Sales
- Total Profit
- Total Quantity
- Total Orders
- Profit Margin
- Monthly Sales Trend
- Monthly Profit
- Sales by Category
- Sales by Sub-Category
- Region filter
- Category filter
- Segment filter
- Order Date filter

## Page 2 — Profitability & Business Insights

- Average Order Value
- YoY Sales %
- Profit by Sub-Category
- Profit by Category
- Sales & Profit by Segment
- Key Business Insights
- Region filter
- Order Date filter

# 09. KPI & Performance Summary

| Total Sales    | \$2.30M   |
|----------------|-----------|
| Total Profit   | \$286.40K |
| Total Quantity | 38K       |
| Total Orders   | 5,009     |

|                     |                       |
|---------------------|-----------------------|
| Profit Margin       | 12.47%                |
| Average Order Value | \$458.61              |
| YoY Sales Growth    | 20.36% (2017 vs 2016) |

## 10. Profit Margin by Category

| Category        | Sales     | Profit    | Profit Margin |
|-----------------|-----------|-----------|---------------|
| Technology      | \$836.15K | \$145.45K | 17.4%         |
| Office Supplies | \$719.05K | \$122.49K | 17.0%         |
| Furniture       | \$742.20K | \$18.45K  | 2.5%          |

## 11. Key Business Insights

**Technology leads profitability:** Technology generated the highest category profit at \$145.45K, with a profit margin of approximately 17.4%.

**Furniture has low profitability:** Furniture generated \$742.20K in sales but only \$18.45K in profit, resulting in a profit margin of approximately 2.5%.

**Phones lead sales:** Phones generated the highest sub-category sales at approximately \$330K.

**Consumer segment leads:** Consumer generated the highest sales at \$1.16M and the highest profit at \$134.12K.

**Copiers lead sub-category profit:** Copiers generated the highest sub-category profit at approximately \$56K.

## 12. Business Recommendations

- **Prioritize high-profit Technology products:** Technology has the highest category profit margin at approximately 17.4%. Focus on high-profit products such as Copiers and Phones to strengthen overall profitability.
- **Review Furniture profitability:** Furniture has a profit margin of approximately 2.5%. Review pricing, discounting, and cost structure to identify opportunities to improve profitability.
- **Leverage Consumer-segment demand:** Consumer is the largest segment by sales and profit. Leverage this strong demand to identify further growth opportunities.

## 13. DAX Measures

### Total Sales

```
Total Sales =
SUM('Sample - Superstore'[Sales])
```

### Total Profit

```
Total Profit =  
SUM('Sample - Superstore'[Profit])
```

## Profit Margin

```
Profit Margin =  
DIVIDE(  
[Total Profit],  
[Total Sales]  
)
```

## Average Order Value

```
Avg Order Value =  
DIVIDE(  
[Total Sales],  
DISTINCTCOUNT('Sample - Superstore'[Order ID])  
)
```

## YoY Sales %

```
YoY Sales % =  
VAR CurrentYear =  
MAX('Sample - Superstore'[Year])  
VAR CurrentSales =  
CALCULATE(  
[Total Sales],  
'Sample - Superstore'[Year] = CurrentYear  
)  
VAR PreviousSales =  
CALCULATE(  
[Total Sales],  
'Sample - Superstore'[Year] = CurrentYear - 1  
)  
RETURN  
DIVIDE(  
CurrentSales - PreviousSales,  
PreviousSales  
)
```

## 14. Project Outcome

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- Data cleaning and preparation
- Exploratory data analysis
- Python-based business analysis
- Sales and profitability analysis
- DAX calculations
- Power BI dashboard development
- KPI development
- Business insight generation
- Business recommendations

The project demonstrates the ability to convert raw business data into an interactive dashboard and communicate analytical findings through business-focused insights.

## 15. Project Structure

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```
E-Commerce-Sales-Profit-Analysis/  
├──  
│   ├── Dataset/  
│   │   └── Sample - Superstore.csv  
│   ├── Python/  
│   │   └── E-Commerce_Analysis.ipynb  
│   ├── PowerBI/  
│   │   └── E-Commerce-Sales-Profit-Analysis.pbix  
│   ├── Dashboard/  
│   │   ├── dashboard-page-1.png  
│   │   └── dashboard-page-2.png  
│   └── README.md
```

## 16. Project Note

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The initial Python-based project workflow was followed from an online tutorial. The project was further developed as a portfolio project by extending the analysis into Power BI, including dashboard development, DAX calculations, business insights, and recommendations.

## 17. Author

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**ADITAYA SINGH**

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SQL • Python • Power BI • Excel

**PORTFOLIO FOCUS**

Business analysis

Interactive dashboards

Data-driven insights

Prepared for professional portfolio presentation.